

ENTREGA FINAL - PROJETO CLOUD PETJOURNEY

Disciplina:

Cloud Computing

Equipe:

PetJourney

Integrantes

- Deryk de Souza Queiroz – RM563412
- Lucas Gonçalves Viana – RM563254
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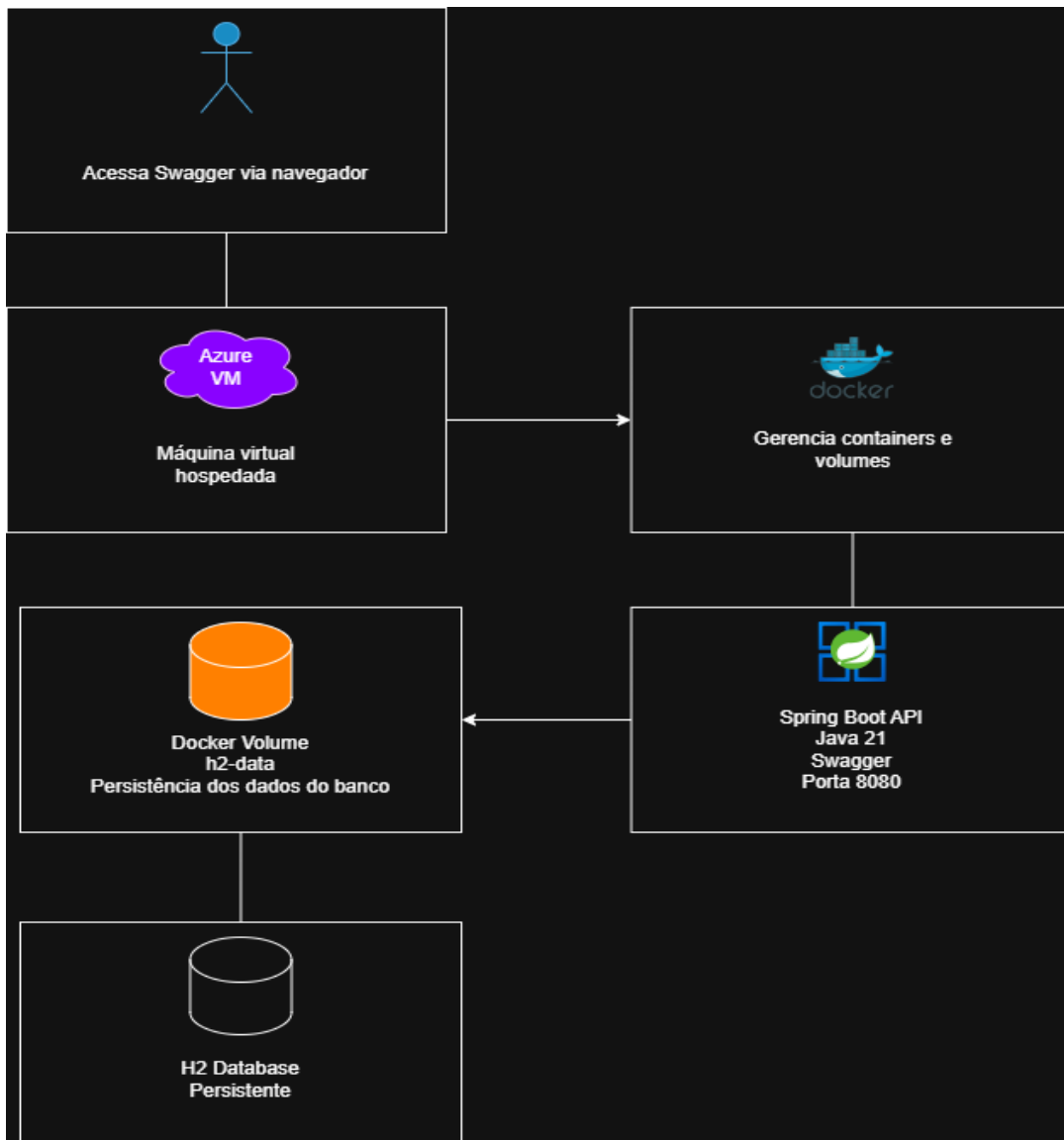
04. Vídeo no YouTube

05. Evidências da infraestrutura

01) Arquitetura da Solução

A solução foi desenvolvida utilizando Java e Spring Boot, containerizada com Docker e publicada em uma máquina virtual Linux na Microsoft Azure.

A arquitetura contempla a execução da API em container Docker, persistência de dados através de volume Docker e exposição da API via porta 8080 utilizando Swagger/OpenAPI.



02) Fluxo da Infraestrutura

- Usuário acessa a aplicação através do Swagger.
- A requisição chega na VM hospedada na Azure.
- O Docker gerencia o container da aplicação Spring Boot.
- A API processa as requisições REST.
- Os dados são persistidos no volume Docker.
- O banco H2 mantém os dados persistentes.

03) Link do Repositório GitHub

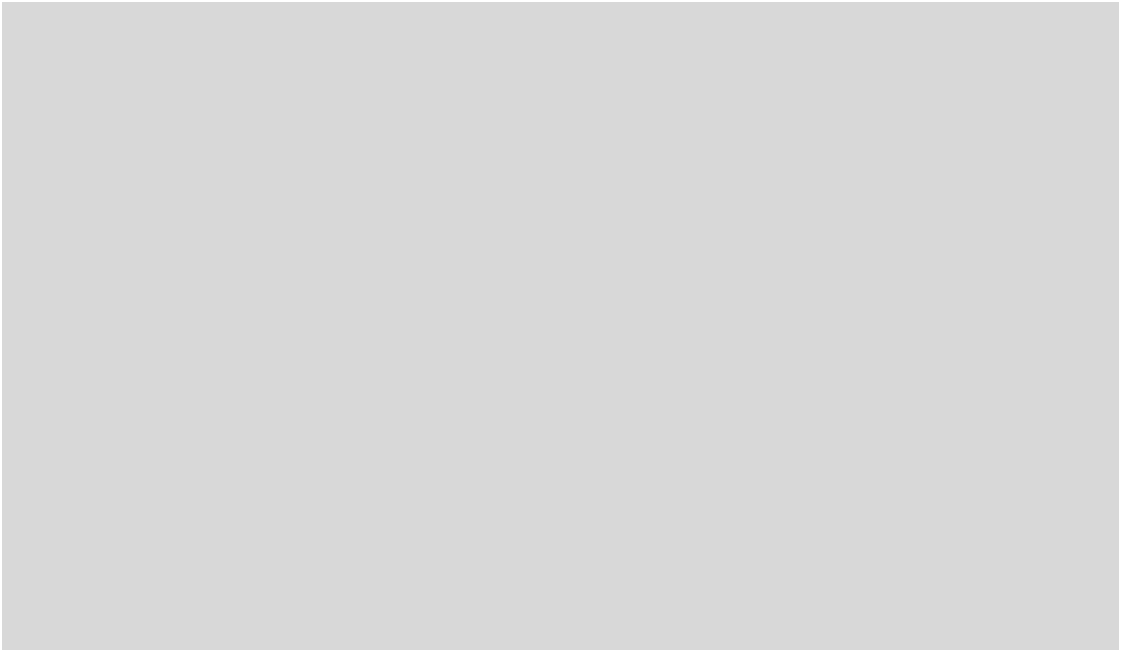
Repositório oficial do projeto:

<https://github.com/LucasViana130/Sprint1-Cloud>

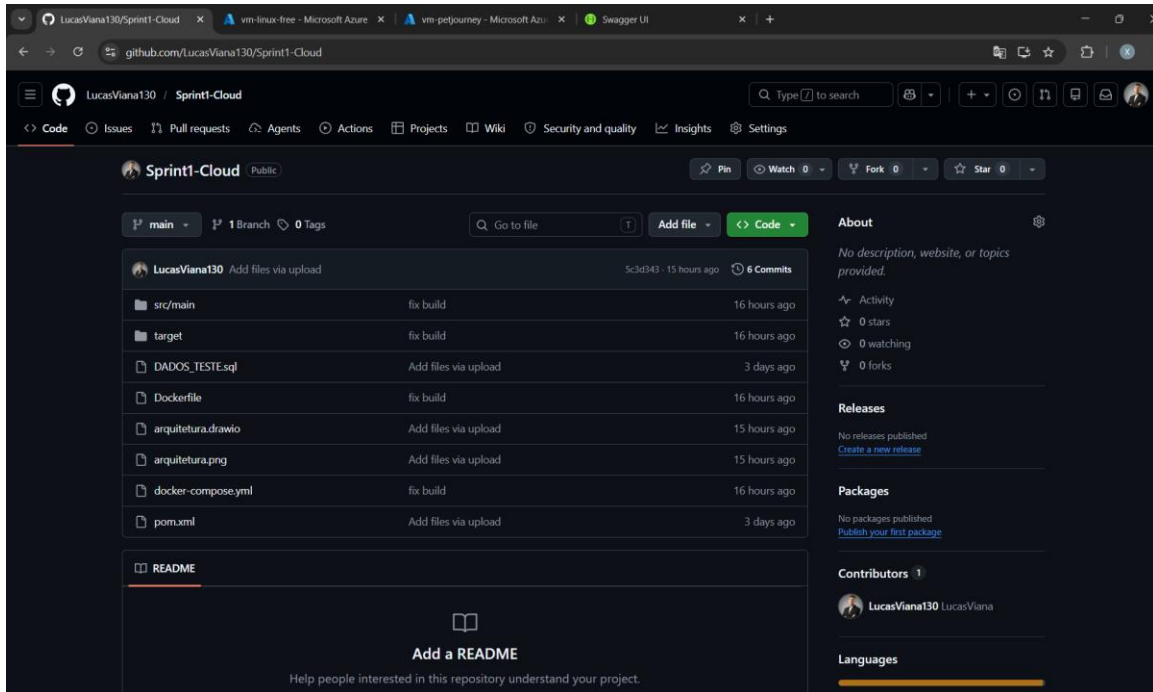
04) Link do Vídeo no YouTube

Link final do vídeo demonstrando a execução do projeto:

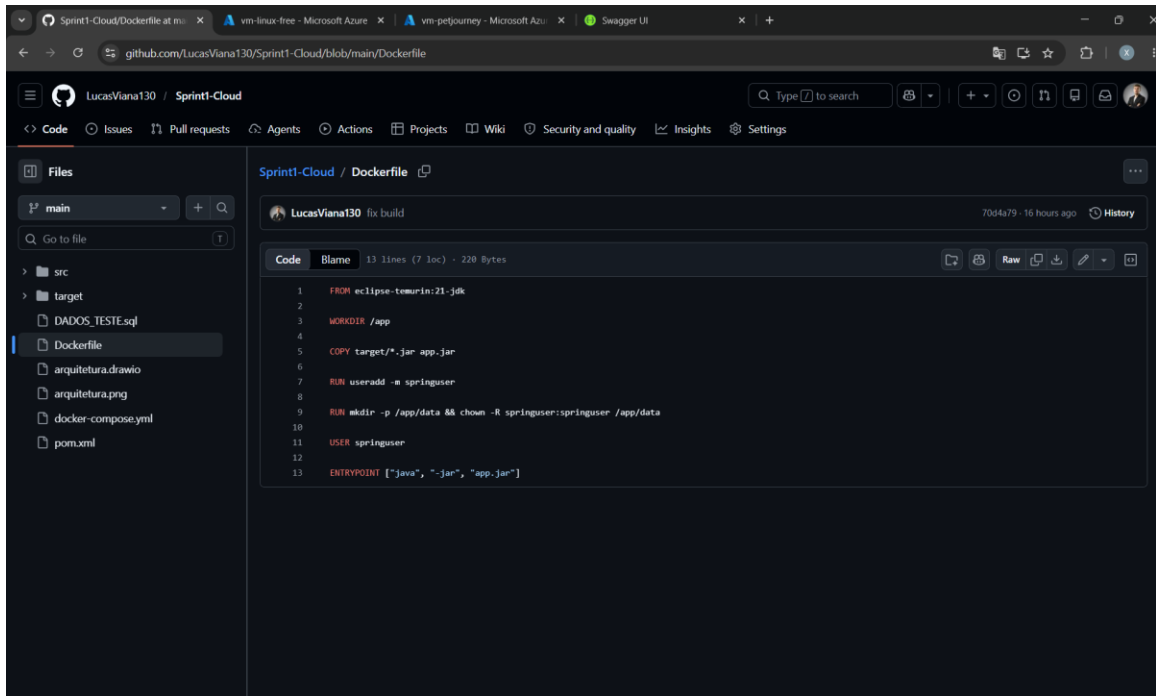
<https://www.youtube.com/watch?v=rRf5zXidzAE>



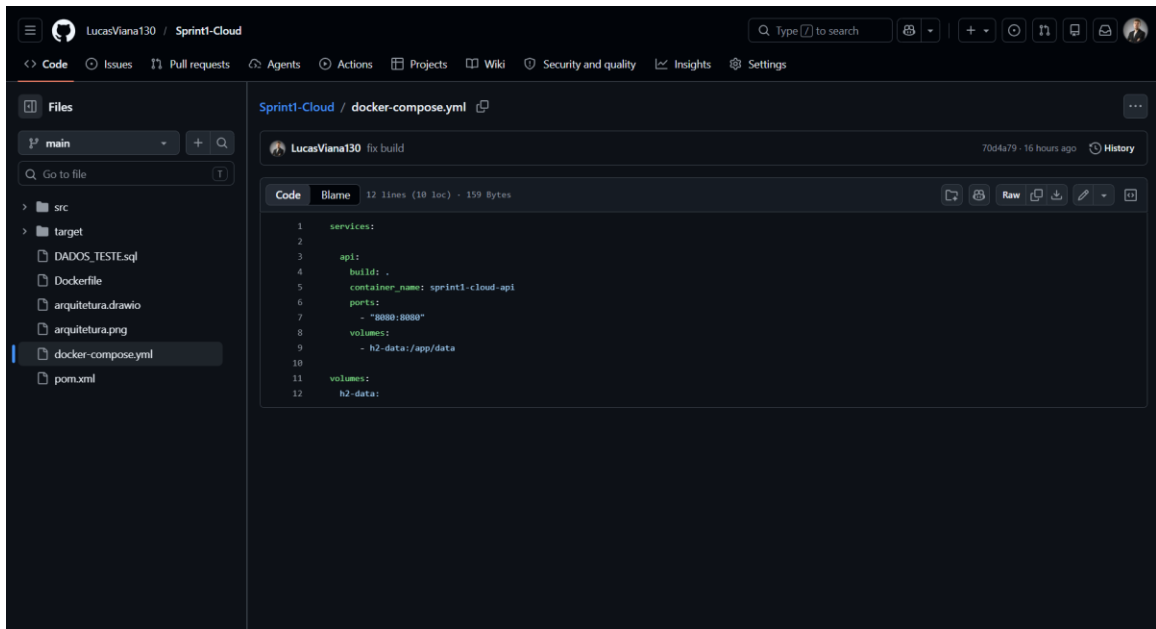
05) Evidências da Infraestrutura



Tela inicial GitHub



DockerFile



Docker-Compose

Screenshot of the Microsoft Azure portal showing the details of a virtual machine named 'vm-petjourney'.

Assistente (1 de 1): Habilite a excelência fundamental do Início Confiável e a segurança moderna para VMs da geração 2 existentes →

Ajude-me a copiar essa VM em qualquer região

Fundamentos

Grupo de recursos (mover)	: tp-petjourney	Sistema operacional	: Linux (almalinux 10.1)
Status	: Em execução	Tamanho	: Standard B2ats v2 (2 vCPUs, 1 GiB de memória)
Local	: Canada Central	IP público da NIC primária	: 4.206.208.16
Assinatura (mover)	: Azure for Students		1 IPs públicos associados
ID da Assinatura	: eec8d38-81e0-46a2-b0b8-d4db69e172ce	Rede virtual/sub-rede	: vnet-canadacentral-1/vnet-canadacentral-1
		Nome DNS	: Não configurado
		Estado de integridade	: -
		Horário criado	: 23/05/2026, 14:58 UTC

Marcações (editar) : Adicionar marcas

Propriedades | Monitoramento | Funcionalidades (7) | Recomendações (1) | Tutoriais

Máquina virtual

Nome do computador	vm-petjourney
Sistema operacional	Linux (almalinux 10.1)
Geração de VM	V2
Arquitetura de VM	x64
Status do agente	Ready
Versão do agente	2.15.2.1
Hibernação	Desabilitado
Grupo de hosts	-
Host	-
Grupo de posicionamento por proximidade	-
Status de Colocalização	N/D
Grupo de reserva de capacidade	-
Tipo de controlador do Azure Data Box Disk	SCSI

Azure Spot

Azure Spot	-
Política de remoção do Azure Spot	-

Disco

Disco de SO	vm-petjourney_OsDisk_1_e9fbf038c0004c3c9932cc1153591ced
-------------	---

Rede

Endereço IP público	4.206.208.16 (Interface de rede vm-petjourney662)
	1 IPs públicos associados
Endereço IP público (IPv6)	-
Endereço IP privado	172.16.0.4
Endereço IP privado (IPv6)	-
Rede virtual/sub-rede	vnet-canadacentral-1/vnet-canadacentral-1
Nome DNS	Configurar

Tamanho

Tamanho	Standard B2ats v2
vCPUs	2
RAM	1 GiB
Threads por núcleo	2

Propriedades | Monitoramento | Funcionalidades (7) | Recomendações (1) | Tutoriais

Máquina virtual

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Azure Spot

Azure Spot	-
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Disco

Disco de SO	vm-petjourney_OsDisk_1_e9fbf038c0004c3c9932cc1153591ced
-------------	---

Rede

Endereço IP público	4.206.208.16 (Interface de rede vm-petjourney662)
	1 IPs públicos associados
Endereço IP público (IPv6)	-
Endereço IP privado	172.16.0.4
Endereço IP privado (IPv6)	-
Rede virtual/sub-rede	vnet-canadacentral-1/vnet-canadacentral-1
Nome DNS	Configurar

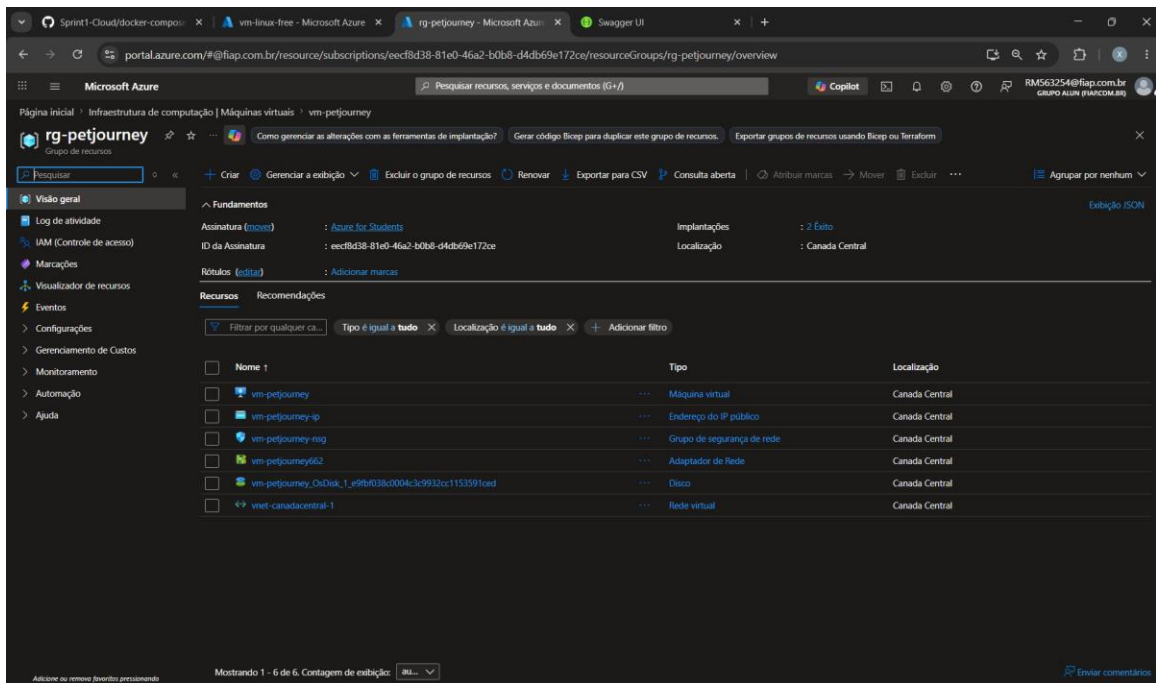
Tamanho

Tamanho	Standard B2ats v2
vCPUs	2
RAM	1 GiB
Threads por núcleo	2

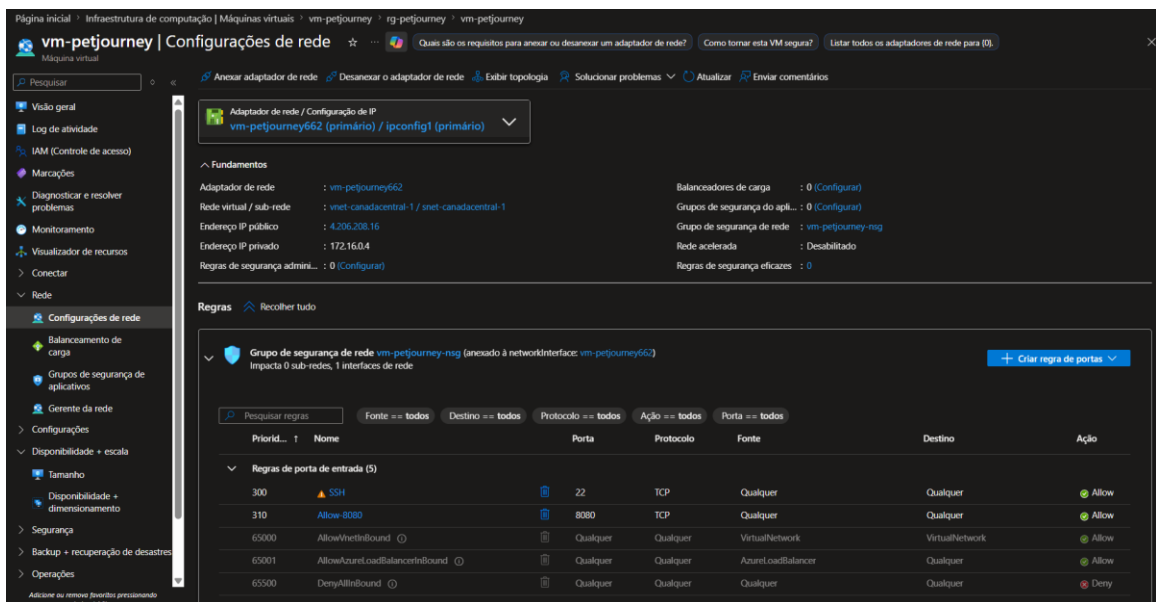
Detalhes de imagens de origem

Fornecedor de imagens de origem	almalinux
Oferta de imagens de origem	almalinux-x86_64
Plano de imagens de origem	10-gen2

Visão geral VM



Grupo de recursos



Configurações de rede

```
portal.azure.com/#@flap.com.br/resource/subscriptions/eecf8d38-81e0-46a2-b0b8-d4db69e172ce/resourceGroups/rg-petjourney/providers/Microsoft.Compute/virtualMachines/vm-...

Microsoft Azure
vm-petjourney | Configurações de rede
Máquina virtual

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
MOID: Share your feedback and help us improve Cloud Shell: https://aka.ms/cloudshell/feedback

VERBOSE: Authenticating to Azure ...
PS /home/lucas> ssh petjourney@4.206.208.16
The authenticity of host '4.206.208.16 (4.206.208.16)' can't be established.
ED25519 key fingerprint is SHA256:r/dZGkFTAc+H3kOwt9LYE/H9gtZLEg99sBmWissGw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '4.206.208.16' (ED25519) to the list of known hosts.
petjourney@4.206.208.16's password:
[petjourney@vm-petjourney ~]$ sudo dnf update -y
```

Configuração SSH realizada na vm

```
portal.azure.com/#@flap.com.br/resource/subscriptions/eecf8d38-81e0-46a2-b0b8-d4db69e172ce/resourceGroups/rg-petjourney/providers/Microsoft.Compute/virtualMachines/vm-...

Microsoft Azure
vm-petjourney | Configurações de rede
Máquina virtual

docker-ce rootless extras-29.5.2-1.el10.x86_64 docker-compose plugin-5.1.4-1.el10.x86_64 nftables-1:1.1.1-9.el10_1.x86_64

complete!
[petjourney@vm-petjourney ~]$ sudo systemctl enable docker
sudo systemctl start docker
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service'.
[petjourney@vm-petjourney ~]$ docker --version
docker version 29.5.2, build 79eb04c
[petjourney@vm-petjourney ~]$ docker compose version
Docker Compose version v5.1.4
[petjourney@vm-petjourney ~]$ docker compose
Usage:  docker compose [OPTIONS] COMMAND

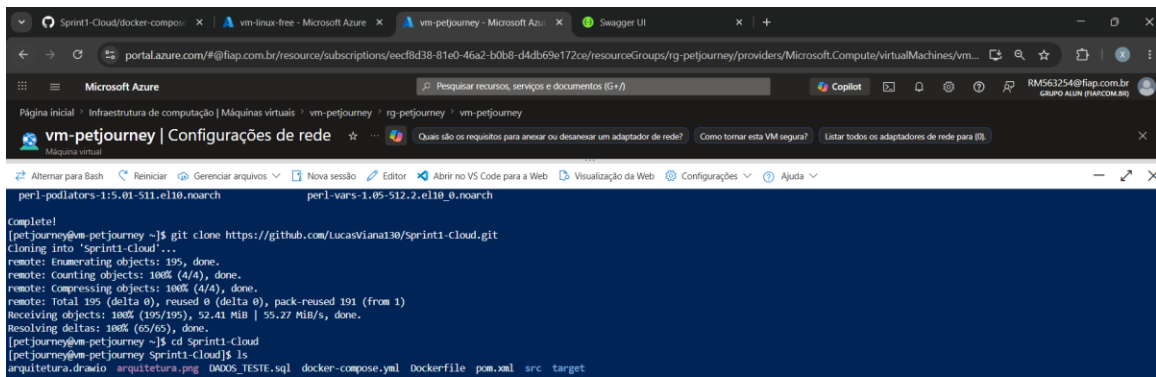
Define and run multi-container applications with Docker

options:
  --all-resources      Include all resources, even those not used by services
  --ansi string        Control when to print ANSI control characters ("never"|"always"|"auto") (default "auto")
  --compatibility       Run compose in backward compatibility mode
  --dry-run            Execute command in dry run mode
  --env-file stringArray Specify an alternate environment file
  -f, --file stringArray Compose configuration files
  --parallel int       Control max parallelism, -1 for unlimited (default -1)
  --profile stringArray Specify a profile to enable
  --progress string    Set type of progress output (auto, tty, plain, json, quiet)
  --project-directory string Specify an alternate working directory (default: the path of the, first specified, compose file)
  -p, --project-name string Project name

Management Commands:
  bridge              Convert compose files into another model

Commands:
  attach              Attach local standard input, output, and error streams to a service's running container
  build               Build or rebuild services
  commit              Create a new image from a service container's changes
  config              Parse, resolve and render compose file in canonical format
  cp                  Copy files/folders between a service container and the local filesystem
```

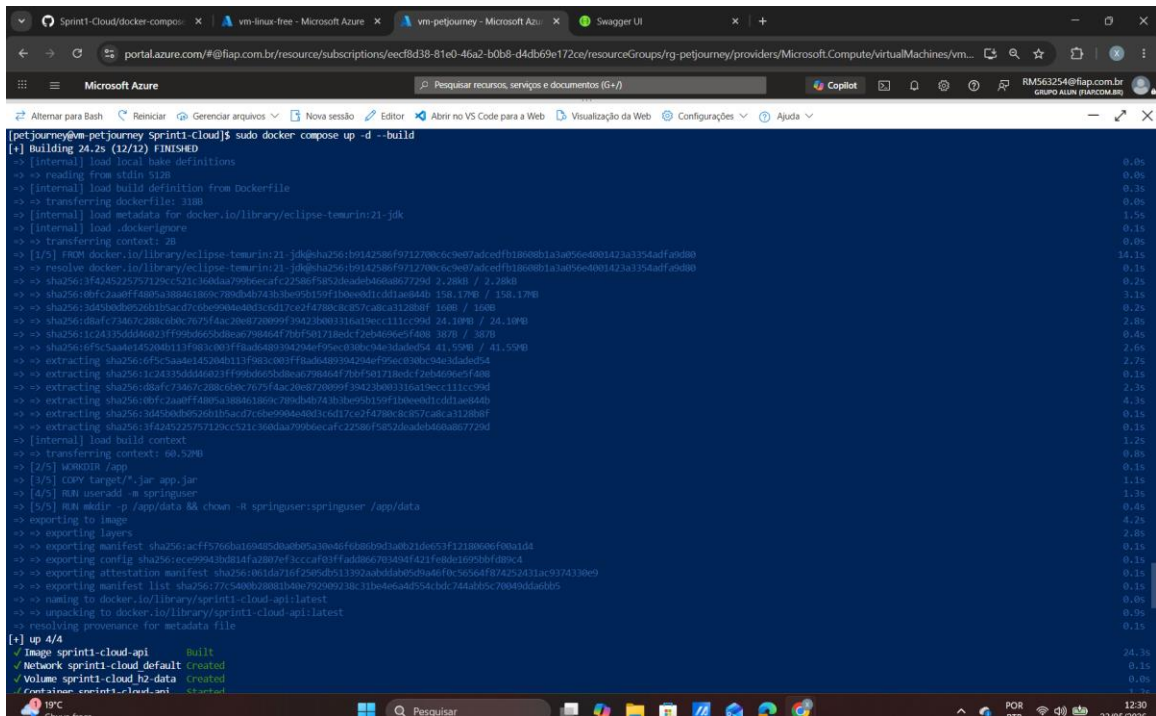
Instalação e configuração do Docker e do Docker Compose na vm



The screenshot shows a terminal window in the Microsoft Azure portal. The user is logged in as 'RM563254@fiap.com.br'. The terminal shows the following commands and output:

```
Complete!  
[pet-journey@vm-pet-journey ~]$ git clone https://github.com/lucasviana30/Sprint1-cloud.git  
Cloning into 'Sprint1-cloud'..  
remote: Enumerating objects: 195, done.  
remote: Counting objects: 100% (4/4), done.  
remote: Compressing objects: 100% (4/4), done.  
remote: Total 195 (delta 0), reused 0 (delta 0), pack-reused 191 (from 1)  
Receiving objects: 100% (195/195), 52.41 MiB / 55.27 MiB/s, done.  
Resolving deltas: 100% (65/65), done.  
[pet-journey@vm-pet-journey ~]$ cd Sprint1-cloud  
[pet-journey@vm-pet-journey Sprint1-cloud]$ ls  
arquitetura.drawio  arquitetura.png  DM005_TESTE.sql  docker-compose.yml  Dockerfile  pom.xml  src  target
```

Clone do repositório na azure



The screenshot shows a terminal window in the Microsoft Azure portal. The user is logged in as 'RM563254@fiap.com.br'. The terminal shows the following commands and output:

```
[pet-journey@vm-pet-journey Sprint1-cloud]$ sudo docker compose up -d --build  
[+] Building 24.2s (12/12) FINISHED  
-> [internal] load local base definitions  
-> -- reading from stdin 0.12s  
-> [internal] load build definition from Dockerfile  
-> -- transferring Dockerfile: 318B  
-> [internal] load metadata for docker.io/library/eclipse-temurin:21-jdk  
-> [internal] load .dockerignore  
-> -- transferring context: 2B  
-> [1/5] FROM docker.io/library/eclipse-temurin:21-jdk@sha256:b9142586f9712708cc9e07adc0df1808b1a3a056e4001423a1354adf5ad80  
-> -- resolve docker.io/library/eclipse-temurin:21-jdk@sha256:b9142586f9712708cc9e07adc0df1808b1a3a056e4001423a1354adf5ad80  
-> -- sha256:3f4245225757129c521c360daa799b6e6cfc22586f5852deadeb46a867729d / 2.28kB  
-> -- sha256:b9142586f9712708cc9e07adc0df1808b1a3a056e4001423a1354adf5ad80 / 158.17kB  
-> -- sha256:245d0b0e22a15c7d4b0e98048d4c0417c9f478bca3877c8b312860f1688 / 16kB  
-> -- sha256:b9142586f9712708cc9e07adc0df1808b1a3a056e4001423a1354adf5ad80 / 24.10kB  
-> -- sha256:1c24355d5d46023f99b0505b2d0a6798665f7bf501718edcf2eb069e5f408 / 387B  
-> -- sha256:6f5c5a4e14520d0113f983c003f8a0d489304204ef95ec030bc94e3da0d54 / 41.59kB  
-> -- extracting sha256:6f5c5a4e14520d0113f983c003f8a0d489304204ef95ec030bc94e3da0d54  
-> -- extracting sha256:1c24355d5d46023f99b0505b2d0a6798665f7bf501718edcf2eb069e5f408  
-> -- extracting sha256:b9142586f9712708cc9e07adc0df1808b1a3a056e4001423a1354adf5ad80  
-> -- extracting sha256:b9142586f9712708cc9e07adc0df1808b1a3a056e4001423a1354adf5ad80  
-> -- extracting sha256:3d45d0d052b1b5ac7c8e990a0d1c6d17c02f4780c8057c8a31280f  
-> -- extracting sha256:3f4245225757129c521c360daa799b6e6cfc22586f5852deadeb46a867729d  
-> [internal] load build context  
-> -- transferring context: 60.52kB  
-> [2/5] WORKDIR /app  
-> [3/5] COPY target/*.jar app.jar  
-> [4/5] RUN useradd -m springuser  
-> [5/5] RUN mkdir -p /app/data && chown -R springuser:springuser /app/data  
-> exporting to image  
-> exporting layers  
-> -- exporting manifest sha256:acf976d3e16948548b6e3a30e4f4eb0b9d1a0b21d0e57f121800e6f0a144  
-> -- exporting config sha256:0ce9943d014fa2807ef3ccaf03f4ad08607034941421f0dd1a092b6dmc4  
-> -- exporting attestation manifest sha256:061d4716f2505d511392abdbab9d8a6f0c56564f074252431ac9374130e0  
-> -- exporting manifest list sha256:77c5400b28081306792909238c31be0e6a6f554c0c744abb5c70040dda0b5  
-> -- naming to docker.io/library/sprint1-cloud-api:latest  
-> -- unpacking to docker.io/library/sprint1-cloud-api:latest  
-> resolving provenance for metadata file  
[+] up 4/4  
Image sprint1-cloud-api Built 24.3s  
Network sprint1-cloud default Created 0.1s  
Volume sprint1-cloud-h2-data Created 0.0s  
Container sprint1-cloud-api Started
```

Build e inicialização da aplicação

```
Sprint1-Cloud/docker-compose: X | vm linux free - Microsoft Azure | vm petjourney - Microsoft Azure | Swagger UI | +
portal.azure.com/#@flap.com.br/resource/subscriptions/eecf8d30-81e0-46a2-b0b8-d4db69e172ce/resourceGroups/rg-petjourney/providers/Microsoft.Compute/virtualMachines/vm...
Microsoft Azure | Pesquisar recursos, serviços e documentos (G+/) | Copilot | RM563254@flap.com.br | GRUPO ALUNO (FRANCONIA.BR)

Alternar para Bash | Reiniciar | Gerenciar arquivos | Nova sessão | Editor | Abrir no VS Code para a Web | Visualização da Web | Configurações | Ajuda

Resolving provenance for includeData file
[+] up 4/4
  Image sprint1-cloud-api Built
  Network sprint1-cloud-default Created
  Volume sprint1-cloud-h2-data Created
  Container sprint1-cloud-api Started
[petjourney@vm-petjourney sprint1-cloud]$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
3e04f9b4c96   sprint1-cloud-api "java -jar app.jar"      11 seconds ago Up 10 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp  sprint1-cloud-api
[petjourney@vm-petjourney sprint1-cloud]$ sudo docker logs sprint1-cloud-api

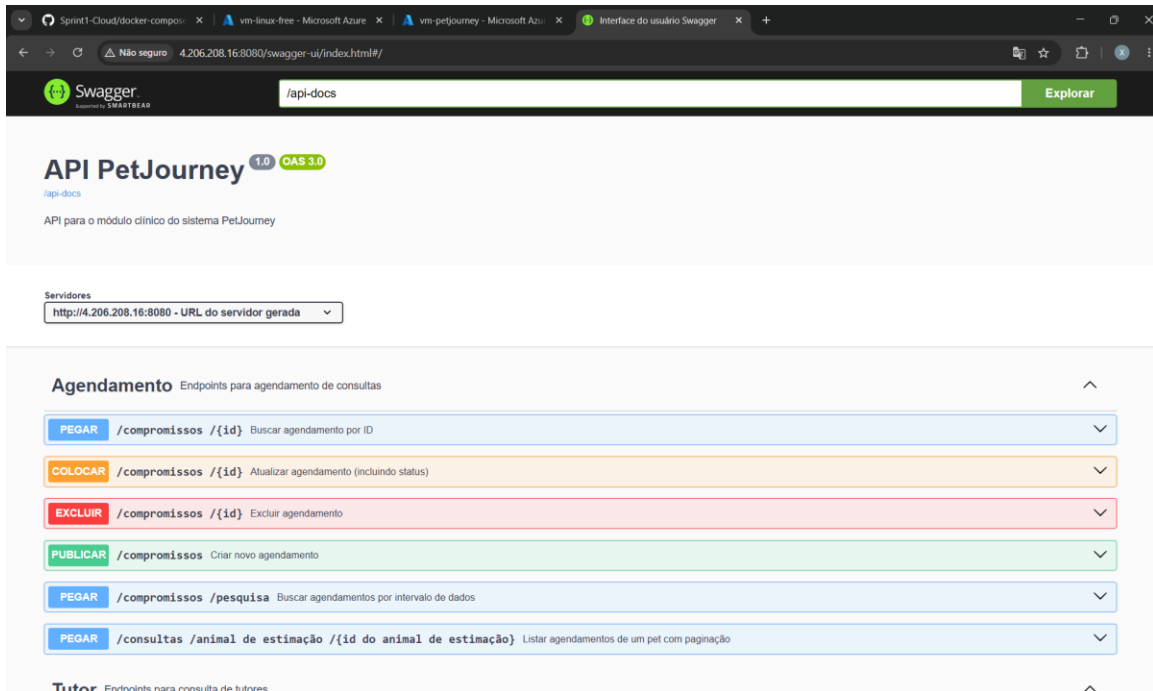
:: Spring Boot ::
(V3.4.0)

2026-05-23T15:08:23.520Z INFO 1 --- [petjourney] [main] b.c.f.petjourney.PetJourneyApplication : Starting PetJourneyApplication v0.0.1-SNAPSHOT using Java 21.0.11 with PID 1 (/app/app.jar started b
y springuser in /app)
2026-05-23T15:08:23.528Z INFO 1 --- [petjourney] [main] b.c.f.petjourney.PetJourneyApplication : No active profile set, falling back to 1 default profile: "default"
2026-05-23T15:08:25.363Z INFO 1 --- [petjourney] [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-05-23T15:08:25.463Z INFO 1 --- [petjourney] [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 88 ms. Found 7 JPA repository interfaces.
2026-05-23T15:08:26.333Z INFO 1 --- [petjourney] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-05-23T15:08:26.353Z INFO 1 --- [petjourney] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-05-23T15:08:26.354Z INFO 1 --- [petjourney] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.33]
2026-05-23T15:08:26.398Z INFO 1 --- [petjourney] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-05-23T15:08:26.400Z INFO 1 --- [petjourney] [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2741 ms
2026-05-23T15:08:26.451Z INFO 1 --- [petjourney] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-05-23T15:08:26.774Z INFO 1 --- [petjourney] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Added connection conn0: url=jdbc:h2:file:./data/challenge user=SA
2026-05-23T15:08:26.777Z INFO 1 --- [petjourney] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-05-23T15:08:26.790Z INFO 1 --- [petjourney] [main] o.s.b.a.h2.H2ConsoleAutoConfiguration : H2 console available at '/h2-console'. Database available at 'jdbc:h2:file:./data/challenge'
2026-05-23T15:08:27.012Z INFO 1 --- [petjourney] [main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-05-23T15:08:27.114Z INFO 1 --- [petjourney] [main] org.hibernate.Version : HHH000042: Hibernate ORM core version 6.6.2.Final
2026-05-23T15:08:27.174Z INFO 1 --- [petjourney] [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-05-23T15:08:27.533Z INFO 1 --- [petjourney] [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-05-23T15:08:27.633Z INFO 1 --- [petjourney] [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 2.3.232
AutoCommit mode: undefined/unknown
```

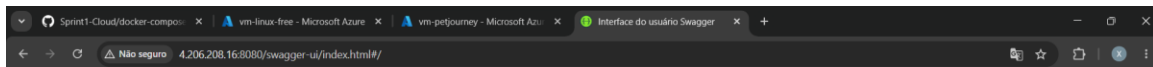
Container Spring Boot em execução com logs de aplicação

```
[petjourney@vm-petjourney sprint1-cloud]$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
62204d2e44a8   sprint1-cloud-api "java -jar app.jar"      11 minutes ago Up 11 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp  sprint1-cloud-api
[petjourney@vm-petjourney sprint1-cloud]$
```

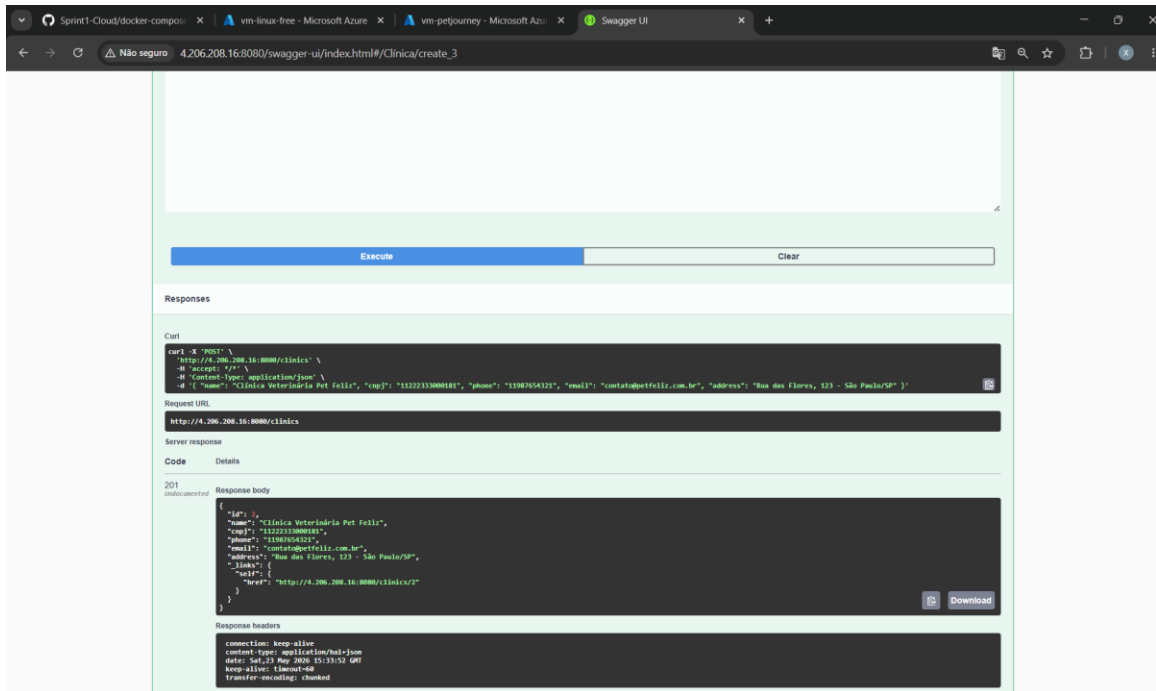
Container Docker da API executando corretamente na porta 8080



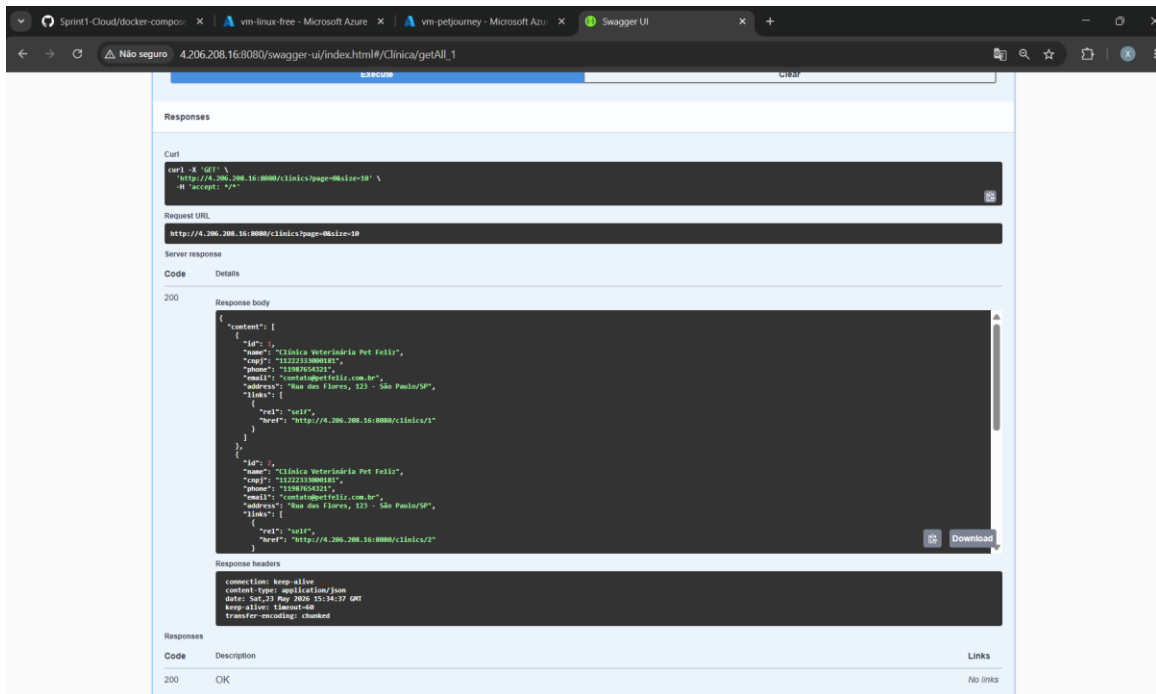
Swagger UI



URL do swagger com ip publico da vm



POST via swagger direto na api



GET via swagger para confirmar o POST